Chao-Wei Huang

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Google Scholar
Github

EDUCATION

National Taiwan University 2018/09 - present
Ph.D. , Computer Science and Information Engineering (CSIE) (expected June 2023)
Advisor: Yun-Nung (Vivian) Chen

National Taiwan University 2014/09 - 2018/06
B. S., Computer Science and Information Engineering (CSIE) GPA: 3.96/4.3

RESEARCH INTERESTS

- Natural Language Processing (NLP)
- Dialogue Systems
- Machine Translation
- Open-domain Retrieval

PUBLICATIONS

[1] Chao-Wei Huang, Tsu-Yuan Hsu, Chen-An Li, Chen-Yu Hsu, Yun-Nung Chen. Unsupervised Multilingual Dense Retrieval via Generative Pseudo Labeling. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (**EACL 2024**)*, Malta, March 17-22, 2024. (**To appear**) [Code]

[2] Chao-Wei Huang, Chen-Yu Hsu, Tsu-Yuan Hsu, Chen-An Li, Yun-Nung Chen. CONVERSER: Few-Shot Conversational Dense Retrieval with Synthetic Data Generation. In *Proceedings of the 24th Annual Meeting of the Special Interest Group on Discourse and Dialogue (**SIGDIAL 2023**)*, Prague, Czechia, September 11-15, 2023. [Code] [Paper]

[3] Tsu-Yuan Hsu*, Chen-An Li*, Chao-Wei Huang, and Yun-Nung Chen. Visually-Enhanced Phrase Understanding. In *Proceedings of The 61st Annual Meeting of the Association for Computational Linguistics (**ACL 2023**)*, Toronto, Canada, July 9-14, 2023. [Paper]

[4] Chao-Wei Huang*, Shang-Chi Tsai*, and Yun-Nung Chen. Modeling Diagnostic Label Correlation for Automatic ICD Coding. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (**NAACL 2021**)*, Online, June 6-11, 2021. [Code] [Paper] [Presentation]

[5] Chao-Wei Huang and Yun-Nung Chen. Learning Spoken Language Representations with Neural Lattice Language Modeling. In *Proceedings of The 58th Annual Meeting of the Association for Computational Linguistics (**ACL 2020**)*, Seattle, USA, July 5-10, 2020. [Code] [Paper] [Slides]

[6] Shang-Yu Su, Chao-Wei Huang, and Yun-Nung Chen. Towards Unsupervised Language Understanding and Generation by Joint Dual Learning. In *Proceedings of The 58th Annual Meeting of the Association for Computational Linguistics (**ACL 2020**)*, Seattle, USA, July 5-10, 2020. [Code] [Paper]

[7] Chao-Wei Huang and Yun-Nung Chen. Learning ASR-Robust Contextualized Embeddings for Spoken Language Understanding. In *Proceedings of 2020 IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP 2020**)*, Barcelona, Spain, 2020. [Code] [Paper] [Slides] [Presentation]

[8] Chao-Wei Huang and Yun-Nung Chen. Adapting Pretrained Transformer to Lattices for Spoken Language Understanding. In *Proceedings of 2019 IEEE Workshop on Automatic Speech Recognition and Understanding (**ASRU 2019**)*, Sentosa, Singapore, December 14-18, 2019. [Code] [Paper]

HONORS & AWARDS

1st Place, Formosa Grand Challenge - Talking to AI 2019/03
The Formosa Grand Challenge(FGC) is a challenge held by the Ministry of Science and Technology. The participants are required to develop a spoken question answering system that is capable of answering questions in spoken content. We won the 1st place overall among over 100 teams.

3rd Place, the 7th Dialogue System Technology Challenge 2018/11
The 7th Dialog System Technology Challenge (DSTC7) provides three challenges for end-to-end dialogue systems. We participated in track 1, dialogue response selection challenge, and ranked 3rd and 4th on two different datasets.

WORKING EXPERIENCES

National Taiwan University
Teaching Assistant

- CSIE5431 Applied Deep Learning (2019 Spring, 2020 Spring, 2021 Spring, 2022 Spring, 2023 Fall)

Meta
Research Scientist Intern 2023/07 - 2023/11
Research on leveraging large language models to end-to-end speech translation. We examined architectural designs and fine-tuning techniques for improved speech-to-text translation.

Research Intern (AI) 2022/09 - 2023/01
Research on evaluating and improving translation accuracy of named entities. We found that nearest neighbor methods improve named entity translation quality.

Amazon
Applied Science Intern, AlexaAI 2021/07 - 2021/10
Research on transfer learning of dialogue state tracking with locale differences and ASR robustness.

Applied Science Intern, AlexaAI 2020/03 - 2020/06
Organization of DSTC9 track 1, focusing on knowledge selection and knowledge-grounded response generation in dialogue systems.

Taiwan AI Labs 2017/07 - 2019/12
Software Engineering Intern, Speech Recognition and Synthesis Group
Developing ASR system that serves as backbone of several products (e.g., Yating Transcriber, Yating Input Method, Yating Pianist). The ASR-assisted products have over a hundred thousand installations.

- Acoustic model data preparation, training and optimization.
- Language model data collection, training and optimization.

TECHNICAL SKILLS

Programming Languages
Python, C++

Deep Learning Frameworks
PyTorch, TensorFlow, Keras

LANGUAGES

Mandarin (native)

English (proficient)